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Simultaneous remagnetization and U–Pb isotope resetting in Neoproterozoic carbonates of the São Francisco craton, Brazil

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Abstract

The southern part of the Neoproterozoic São Francisco basin, in Minas Gerais State, Brazil, can be divided into three structural domains: (a) the central part of the basin where the rocks are undeformed; (b) the western domain where the rocks have been deformed by the 600–550 Ma Brasília fold belt; and (c) the eastern domain where the rocks have been affected by the 600–550 Ma Araçuaí fold belt. U–Pb and Pb isotopic data, rock magnetism data and paleomagnetism data from the carbonates from different domains support a close connection between a pervasive remagnetization and a large scale fluid percolation event that strongly affected the isotopic system of these rocks at 530–500 Ma, during the last stage of the Brasiliano/Pan-African orogeny. A Pb–Pb isochron age of 686 ± 69 Ma has been determined from undeformed carbonates in the center of the basin, and it is interpreted as the minimum depositional age. However, most of the Pb–Pb and the U–Pb ages obtained from deformed as well as other undeformed carbonates fall in an interval of 550–500 Ma. Carbonates containing radiogenic crustal Pb with an isotopic signature of the Archean/Paleoproterozoic basement were found in the central portion of the basin, which was not affected by deformation, suggesting that this Pb was incorporated into the carbonates through fluids which promoted the resetting of the isotope system and severe remagnetization in the carbonates. The post-depositional character of the characteristic magnetizations is strengthened by the following: (a) the disclosed rock magnetic properties, such as wasp-waisted hysteresis loops, anomalously high hysteresis ratios and contradictory Lowrie–Fuller and Cisowski tests, are typical of remagnetized carbonates; (b) thermomagnetic analysis and scanning electron microscopy suggest authigenic magnetite as the main magnetic carrier; (c) moderate to high paleolatitudes inferred from paleomagnetic data for the study area would require a different climate pattern during the sedimentation of the wide carbonate platforms; (d) magnetization directions with a single polarity were found along the whole sedimentary sequence; (e) mean magnetization components identified in the carbonates show lower dispersion than would be expected if the secular variation of the geomagnetic field was fully averaged out; (f) paleomagnetic poles from carbonate sequences and adjacent Brasiliano metamorphic rocks are similar and coincide with high quality Gondwanan paleomagnetic poles for the 530–500 Ma interval. The similarity between paleomagnetic and isotopic results from the Bambuí and the Salitre carbonates ca. 1000 km to the northeast implies a large scale fluid percolation event that simultaneously affected the whole basin. The paths of these fluids may have been along old basement faults reactivated during the last stage of the Brasiliano/Pan-African orogeny. © 2000 Elsevier Science B.V. All rights reserved.

Keywords: Carbonates; Neoproterozoic; Paleomagnetism; Remagnetization; U–Pb geochronology

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1. Introduction

Carbonates have been broadly used in paleomagnetic studies. Although some examples of primary magnetization acquired during deposition or diagenesis have been reported (e.g. Channell and McCabe, 1994; Tarduno and Myers, 1994; Belkaaloul and Aïssaoui, 1997), pervasive remagnetization seems to be frequent in these rocks (e.g. McCabe and Elmore, 1989; Jackson, 1990; Jackson et al., 1993; McCabe and Channell, 1994; Huang and Opdyke, 1996). These remagnetized carbonates are easily recognized by their typical magnetic properties, such as wasp-waisted hysteresis loops, anomalously high hysteresis ratios and contradictory Lowrie–Fuller and Cisowski tests (Jackson, 1990; McCabe and Channell, 1994; Channel and McCabe, 1994; Huang and Opdyke, 1996). Rock magnetic and electron microscopy studies in Paleozoic carbonates from North America suggest authigenic magnetite as the main carrier of secondary chemical remanence (Suk et al., 1990; Jackson, 1990), probably formed by migration of warm and chemically active orogenic fluids toward the cratonic basin (Miller and Kent, 1988; Jackson et al., 1988; McCabe et al., 1989). Intriguingly, these strongly remagnetized limestones yield depositional U–Pb ages, suggesting that the isotopic system was not affected by the remagnetization event (DeWolf and Halliday, 1991).

Mobilization and migration of fluids during orogenic events have been widely recognized (e.g. Oliver, 1986) and are considered responsible for Mississippi Valley Type deposit genesis (e.g. Hearn and Sutter, 1985; Leach and Roman, 1986; Duane and DeWit, 1988), formation of authigenic K-feldspar (Hearn and Sutter, 1985), diagenesis of clay minerals and petroleum migration (Dickinson, 1974; Morton, 1985).

In Brazil extensive Neoproterozoic carbonatic sequences of the São Francisco basin cover $>250\,000\ km^2$ of the western part of the São Francisco craton (Fig. 1). The São Francisco basin is partially affected by the encircling Brasiliano orogenic belts (Almeida, 1967) and present Pb–Zn mineralizations of Mississippi Valley type. The Brasiliano orogeny was defined as a long-lived cycle [900–500 Ma; Brito Neves and Cordani (1991)] correlated to the Pan-African orogeny in Africa. However, some recent works have divided these orogenies in two stages with narrow age intervals (e.g., Meert et al., 1995; Trompette, 1997). The first stage (900/800–600 Ma) is related to the Adamastor and Mozambique oceans closure, and peaked at $\sim$600 Ma. The second stage (550–500 Ma) is related to a widespread set of radiometric ages found throughout the Gondwana supercontinent. It has been regionally called the Ross Orogeny in Australia and the Cape orogeny in South Africa. In South America this stage has not yet been named and is referred in this paper as the last stage of the Brasiliano orogeny.

Paleomagnetic and geochronological studies have been carried out in the carbonates of the São Francisco basin (e.g. D'Agrella-Filho, 1995; D'Agrella-Filho et al., 1997; Thomaz-Filho et al., 1998; Misi and Veizer, 1998; Babinski et al., 1999). The depositional age of the carbonates is still disputable since results from the various isotopic dating methods led to conflicting interpretations. Rb–Sr ages on clays and whole-rock samples range from 695 ± 12 Ma to 465 ± 21 Ma and K–Ar ages on clays range from 576 ± 14 Ma to ca. 478 Ma (Thomaz-Filho et al., 1998 and references therein). Recently, Kawashita et al. (1997) and Misi and Veizer (1998) proposed a sedimentation age of ca. 600 Ma for the carbonatic sequences of the Bambuí and Una Groups (Fig. 1) based on $^{87}Sr/^{86}Sr$ ratios. On the other hand, Babinski et al. (1999), based on Pb isotope data, consider 690 Ma as the minimum age for the Bambuí sedimentation and recognized a widespread resetting event at 550–500 Ma.

Our main goal is to show by means of rock magnetism, paleomagnetism, and U–Pb and Pb isotope data the close connection between a pervasive remagnetization and a large scale fluid percolation event which strongly affected the isotopic system in the carbonates at about 530–500 Ma, during the second stage of the Brasiliano/Pan-African orogeny.

2. Geologic setting and sampling

The São Francisco basin is a Neoproterozoic sedimentary sequence, the São Francisco

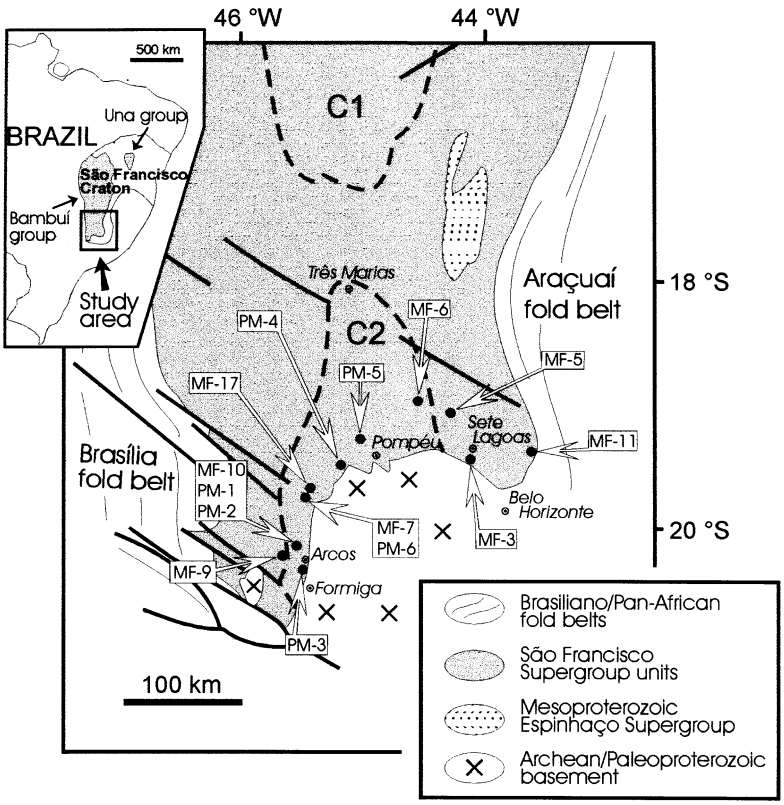


Fig. 1. Southern São Francisco Basin geological map [modified from Chemale et al. (1993)]. C1 and C2 represent stable (undeformed) cratonic areas not affected by the Brasiliano orogeny. Sampling localities are indicated by full circles (MF-geochronological and PM-paleomagnetic sampling localities). The inset map shows the São Francisco Supergroup units (Bambuí and Una Groups).

Supergroup, composed of two groups (Dardenne, 1978): the basal Macaúbas Group (and correlative units, Bebedouro and Jequitaí Formations) which represents a widespread glacial record and the overlying Bambuí Group (and correlative unit, Salitre Formation). The Bambuí Group consists of two main sequences: basal shallow-water marine strata, comprising two cycles of carbonate and pelitic–psamitic sedimentation and an upper shallow-water to alluvial strata. The Bambuí carbonates host Pb–Zn mineralizations that are considered to be Mississippi Valley type deposits. The mineralizations occur in fractures, fault planes and crests of folds in different localities of the basin, indicating that they formed after the deposition of the carbonates. Deformation in these rocks increases progressively towards the enclosing fold belts: the east-verging Brasília fold belt and the west-verging Araçuaí fold belt. Metamorphism grades from greenschist facies at the borders to anchimetamorphism in the center of the basin. Two undeformed regions (Fig. 1) were recognized in the center of the basin (Chemale et al., 1993).

For this study carbonate rocks from the southern São Francisco basin were sampled, comprising mainly the basal unit of the Bambuí Group (Sete Lagoas Formation). Detailed sampling for Pb isotope study was carried out on 8 outcrops, from regions with different degrees of deformation (Fig. 1); these rocks were analyzed for Pb isotopic compositions and U and Pb concentrations. Four outcrops are located on the stable area of the basin (MF-6, MF-7, MF-10 and MF-17); three outcrops (MF-3, MF-5 and MF-11) are from the area where the rocks have been affected by the Araçuaí fold belt and one outcrop (MF-9) is located on the

area affected by the Brasília fold belt. For paleomagnetic purposes 46 nearly-horizontal stratigraphic layers (46 sites) of limestones and carbonatic shales were sampled at four quarries (PM-1, PM-2, PM-5 and PM-6; Fig. 1); at outcrocrops along a hill (shales) close to the Arcos-Formiga road (PM-3), and at an outcrop along a road in the Pompéu region (PM-4), comprising >100 m in the sedimentary sequence from the undeformed area. Most of these localities are coincident with those sampled for isotopic studies (Fig. 1). Six or seven cylinders were extracted from each stratigraphic layer using a portable gasoline-powered drill; whenever possible, sun and magnetic compasses were used for orienting samples.

3. Rock magnetic fingerprints of remagnetization

The hysteresis properties of the rocks studied are typical of remagnetized carbonates. Hystereses loops performed for 38 samples using a Molspin-VSM apparatus show a wasp-waisted shape (Fig. 2a), which is characteristic of a bimodal distribution of soft and hard coercivity fractions. The saturation remanence to saturation magnetization (J_{rs}/J_s) ratio against remanent coercivity to bulk coercivity (H_{cr}/H_c) ratio diagram (after Day et al., 1977) is shown in Fig. 2(b). The hysteresis ratios are distributed throughout the pseudo-single-domain (PSD) and multidomain (MD) fields and show anomalously high H_{cr}/H_c values, plotting along the Jackson (1990) trend for remagnetized carbonates of North America.

To further aid in the magnetic mineralogy characterization, the modified form of Lowrie and Fuller test (Lowrie and Fuller, 1971; Johnson et al., 1975), the Cisowski (1981) test and the Lowrie (1990) test were performed on samples from 10 representative sites. They were first stepwise demagnetized in alternating field (AF) up to 200 mT, and then given an anhysteretic remanent magnetization (ARM) in a peak AF of 200 mT and a biasing field of 0.1 mT. The ARM was then stepwise AF demagnetized. Finally, the samples were given an incremental isothermal remanent magnetization (IRM) using a SI-4 AF demagnetizer (Sapphire Instruments) and were then progressively AF demagnetized [Fig. 3(a)]. Additionally, samples from the same sites were thermally demagnetized after isothermal remanence acquisition along three orthogonal directions with different fields (1.4, 0.4 and 0.12 T) to estimate unblocking temperatures of the soft (<0.12 T), medium (0.12–0.4 T) and hard (>0.4 T) coercivity fractions [Fig. 3(b)].

The Lowrie–Fuller and Cisowski tests [Fig. 3(a)] show contradictory results. The IRM acquisition and demagnetization curves are symmetrical and cross over at near 50% of the normalized remanence, which suggest that the remanence predominantly resides in non-interacting fine (SD or PSD) particles according to Cisowski (1981). On the other hand, ARM is weaker than IRM to AF demagnetization, suggesting that the remanence carrier is coarse grained (MD) magnetite. This apparent contradictory behavior has also been observed in the remagnetized Paleozoic carbonates from North America (Jackson, 1990), England (McCabe and Channell, 1994) and Southwest China (Huang and Opdyke, 1996). Unremagnetized Mesozoic limestones from Italy and Middle Jurassic carbonates from the Paris basin exhibit a different behavour showing comparable ARM and IRM stabilities against AF demagnetization (Channell and McCabe, 1994; Belkaaloul and Aïssaoui, 1997).

For some samples [e.g. sample BI-27, Fig. 3(b)] the Lowrie test indicates a hard fraction with unblocking temperatures up to 340°C and medium and soft fractions with unblocking temperatures up to 500°C. The soft and medium IRM components could be correlated to magnetite and the hard IRM component could indicate the presence of sulphides (pyrrhotite). For other samples [e.g. sample BI-49, Fig. 3(b)], however, the hard fraction is also associated with high unblocking temperatures (up to 500°C). Thermomagnetic curves (performed under Ar-atmosphere in a Kappabridge-CS3 apparatus) show a strong decrease in magnetic susceptibility at ca. 580°C (Fig. 4). They also show an irreversible behavior suggesting mineralogical transformation during heating which could be explained by magnetite formation from sulphides (Dekkers, 1990). These results together with hysteresis data point to either

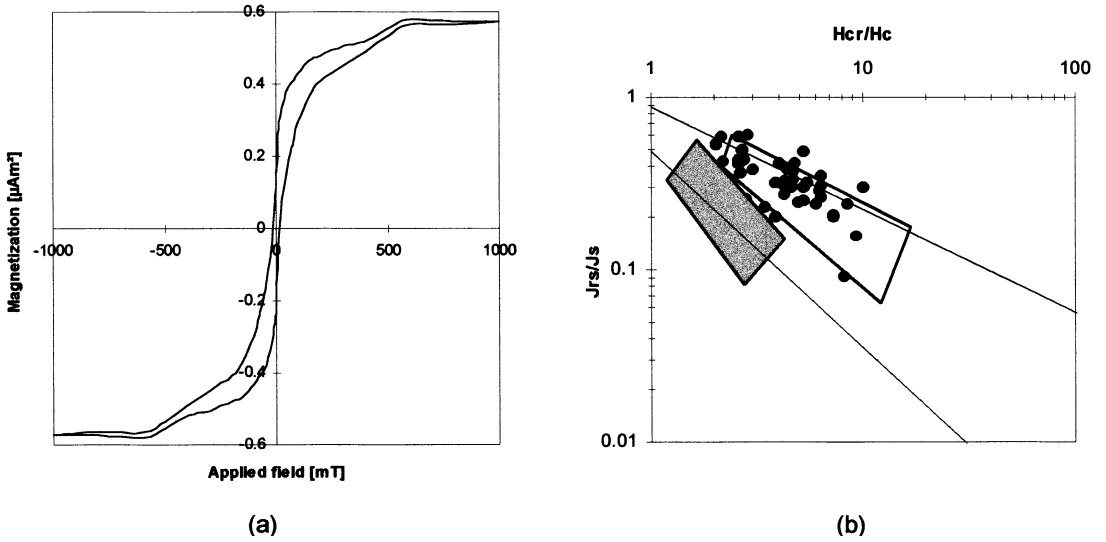


Fig. 2. (a) Typical hysteresis loop showing the 'wasp-waisted' shape; (b) logarithmic plot of hysteresis parameters (J_{rs}, remanence of saturation; J_s, saturation magnetization; H_{cr}, remanent coercivity; H_c, coercive force). Full line represents the trend for North American remagnetized limestones (Jackson, 1990). White quadrilateral delimits the hysteresis results from remagnetized North America and Craven Basin (Great Britain) limestones according to Channell and McCabe (1994). The dashed line represents the trend for SD and MD mixtures from synthetic samples (Parry, 1982) and is similar to the trend defined for preserved primary magnetization rocks (Tarduno and Myers, 1994). The shaded quadrilateral delimits the hysteresis results from the unremagnetized Maiolica limestones (Italy), according to Channell and McCabe (1994).

different grain-size fractions of magnetite or different magnetic minerals (magnetite and sulphides) in the carbonates.

Preliminary results from scanning electron microscopy (SEM) and energy dispersive spectrometer (EDS) indicate spherical magnetites 1–5 μm in diameter and irregular grains of magnetite up to 10 μm long, usually as void filling aggregates. Some magnetites replace pyrite grains. These features were also observed in carbonates from North America and are indicative of an authigenic origin for these magnetite grains (Suk et al., 1990; Sun and Jackson, 1994).

4. Paleomagnetic results and correlations

Both AF and thermal demagnetizations were applied to 495 cylindrical specimens (2.2 cm × 2.5 cm) to separate the magnetic components in the rocks. Samples from eight sites displayed magnetization intensities close to the noise level of the 2G-cryogenic magnetometer, and the remanence vectors showed random paths during demagnetization. For the remaining 38 sites, thermal demagnetization was more efficient than AF treatment in separating magnetization components.

The samples showed a multicomponent magnetic behavior. Two to four magnetic components for each analyzed specimen were identified using the least squares fit method (Kirshvink, 1980) [Fig. 5(a)]. The heating up to 150°C eliminated a magnetization component very close to the present geomagnetic field, interpreted as a viscous magnetization. After 150°C heating, most samples yielded two more components: the component A, a northern, high negative inclination direction, with unblocking temperatures between 150°C and 250/275°C, and the component B, a northeast, high positive inclination direction with unblocking temperatures in the 250/300–400°C range. For 17 sites the slightly different component C was identified for temperatures usually between 340/360°C and 530°C. Fisher's (1953) statistics was used to calculate site mean directions for the A, B and C components [Fig. 5(b); Table 1].

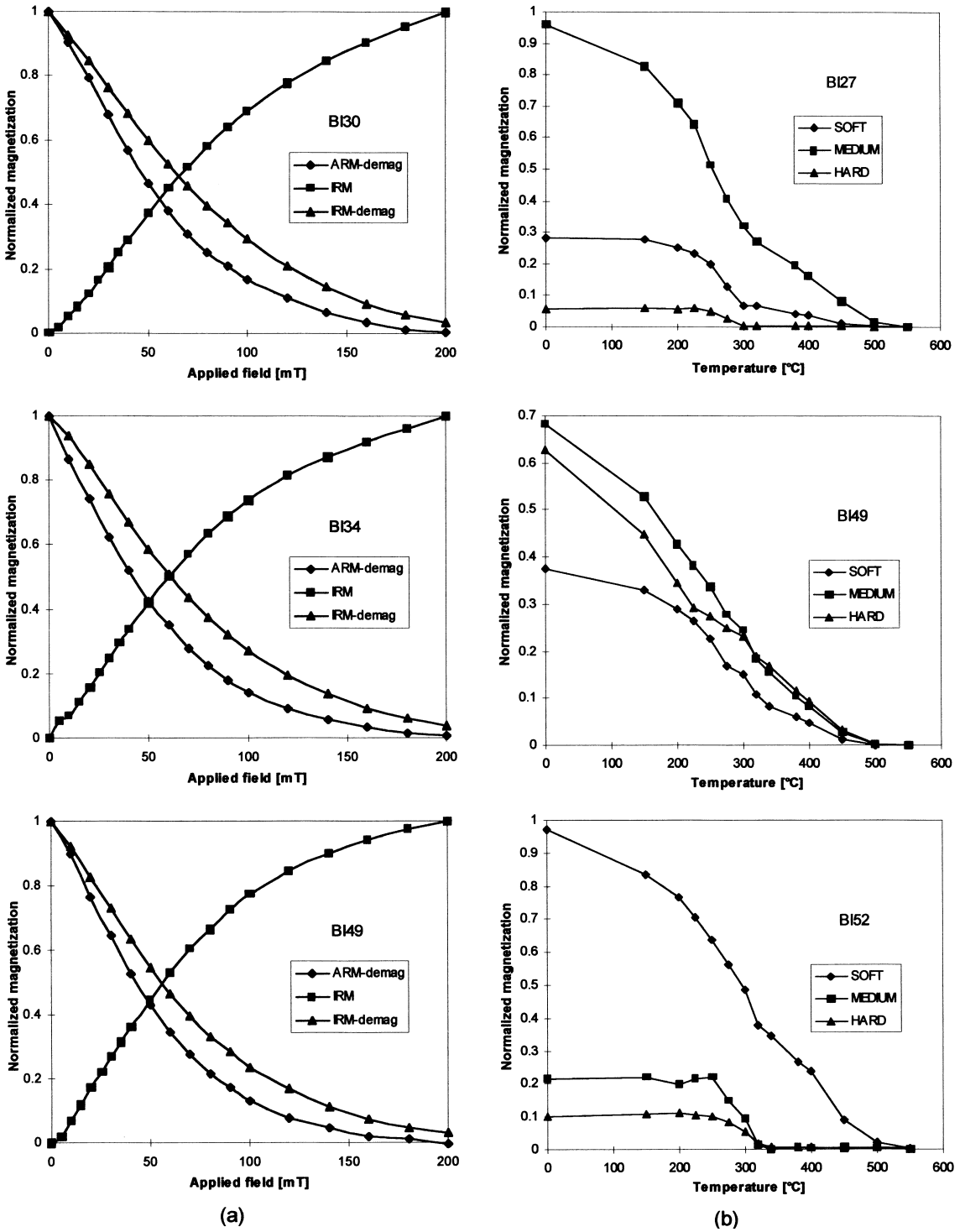


Fig. 3. (a) Examples of Cisowski (1981) and modified Lowrie–Fuller (Johnson et al., 1975) tests: acquisition and AF demagnetization of IRM, and AF demagnetization of ARM (acquired in a peak AF of 200 mT and a biasing field of 0.1 mT); (b) examples of the Lowrie (1990) test: thermal demagnetization after isothermal remanence acquisition along three orthogonal directions with different fields (1.4, 0.4 and 0.12 T) to estimate unblocking temperatures of the soft (<0.12 T), medium (0.12–0.4 T) and hard (>0.4 T) coercivity fractions.

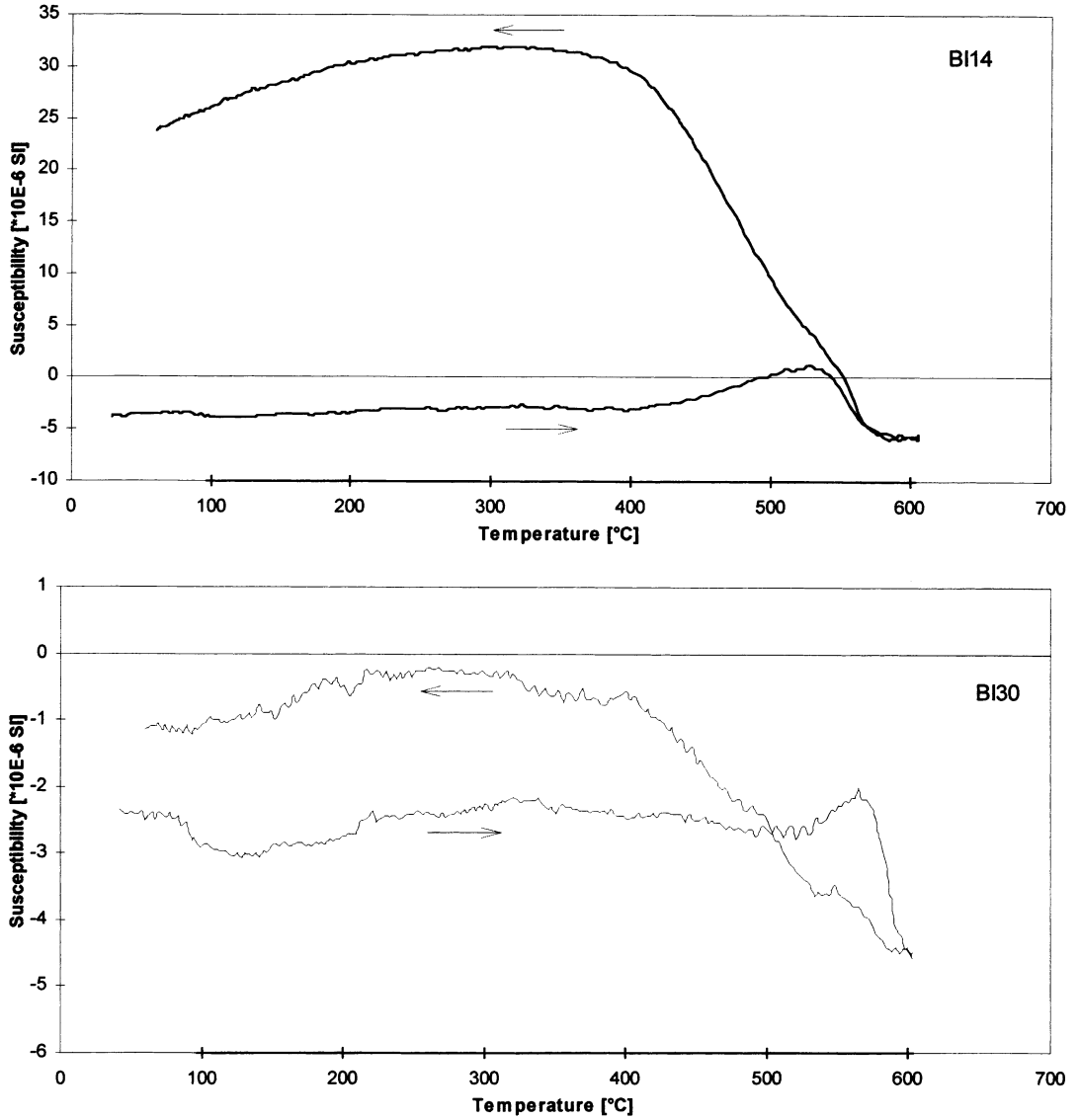


Fig. 4. Magnetic susceptibility versus temperature curves (performed under Ar-atmosphere in a Kappabridge-CS3 apparatus).

The A component has an unconstrained age, although it might have been acquired at the end of Paleozoic times (Early Permian), as suggested by the calculated paleomagnetic pole when compared to other Phanerozoic South American poles (Rapalini and Tarling, 1993, Rapalini et al., 1993). Their low unblocking temperatures is consistent with a remagnetization of thermoviscous origin. The B and C components show higher unblocking temperatures and high stability during thermal demagnetization [Fig. 5(a)]. Correlation with the Lowrie test [Fig. 3(b)] indicates that grains with hard and medium coercivities are the main carriers of remanent magnetization, as already suggested by Sun and Jackson (1994) for remagnetized carbonates from North America. This is clearly observed in sample BI-52 [Fig. 3(b)] for which the hard and medium fractions have unblocking tem-

Table 1
Components A, B and C disclosed for the analyzed samples[a]

Site	*N*	Mean direction				VGP	
		D (°)	*I* (°)	α_{95}(°)	κ	Long. (°)	Lat. (°)
Component A							
7	12	355.9	−59.7	6.3	48.9	143.3	69.5
8	11	359.9	−61.0	4.0	129.8	134.6	68.3
9	8	6.2	−65.3	6.3	77.6	125.3	62.5
10	9	356.0	−62.5	2.4	445.7	141.6	66.3
11	6	5.7	−63.1	5.3	162.9	124.7	65.3
12	9	5.2	−63.8	4.0	166.3	125.9	64.5
13	4	349.2	−64.8	5.0	335.4	150.4	62.2
14	9	358.1	−59.2	6.8	58.4	138.8	70.3
15	4	360.0	−51.5	9.5	93.9	134.6	78.3
16	6	324.8	−53.6	14.5	22.3	193.2	56.1
23	10	352.0	−61.9	5.2	86.4	148.9	65.8
24	7	358.9	−66.2	7.8	60.3	136.3	61
25	10	359.9	−73.1	5.8	70.2	134.9	50.9
26	6	3.3	−61.1	8.5	196.6	128.5	67.2
27	10	345.1	−79.3	10.2	23.5	141.5	39.4
29	12	354.6	−69.1	4.6	88.3	140.6	56.9
30	10	0.7	−72.7	5.4	82.2	134.0	51.7
31	11	348.3	−62.9	2.7	297.4	153.9	64.1
32	12	1.7	−68.3	5.3	68.6	132.4	58.9
33	13	358.5	−66.8	3.0	197.2	136.5	61.0
34	10	0.8	−63.4	4.3	129.7	133.1	65.4
35	12	1.1	−61.4	3.0	206.2	132.3	67.8
36	6	24.7	−59.8	17.6	15.5	94.7	60.4
37	13	349.4	−64.6	2.5	272.5	150.4	62.5
38	11	358.4	−64.6	3.3	188.4	137.0	63.9
39	11	355.5	−56.3	4.3	114.5	146.9	73.1
40	11	2.1	−63.4	4.2	121.8	130.9	65.4
48	11	2.3	−59.2	3.1	222.0	129.2	70.3
49	11	353.3	−65.8	4.8	92.0	143.9	61.8
50	6	358.2	−62.5	5.8	135.8	137.7	66.5
52	7	10.6	−68.2	7.2	71.6	122.0	57.9
Component B							
7	13	23.5	64.8	3.7	127.1	331.3	19.7
8	11	6.6	67.2	3.5	174.6	318.9	19.5
9	13	37.7	65.2	4.3	94.0	339.8	14.4
10	12	26.1	61.0	2.8	235.7	335.2	23.2
11	13	23.1	68.3	3.0	188.4	329.1	15.4
12	10	20.3	67.5	2.4	405.8	327.8	17.1
13	8	39.4	70.6	4.6	145.8	336.1	7.7
14	12	13.3	63.6	3.9	127.7	324.6	23.4
15	12	12.7	61.8	3.9	122.9	324.8	25.6
16	8	21.9	67.0	5.9	88.9	329.2	17.4
21	7	31.2	72.0	6.4	89.1	331.2	9.0
22	9	38.3	68.0	2.9	313.0	338.0	11.6
23	13	26.4	69.3	3.1	175.1	330.6	13.8
24	7	30.4	73.3	4.1	217.5	330.0	7.6
25	8	44.3	73.3	4.9	127.8	335.9	3.5
26	12	13.0	73.5	2.8	240.9	321.5	10.4
27	11	35.7	67.6	2.6	310.3	337.2	13.3

Table 1 *continued*

Site	*N*	Mean direction				VGP	
		D (°)	*I* (°)	α_{95}(°)	κ	Long. (°)	Lat. (°)
29	12	30.8	72.9	2.2	389.7	330.3	7.8
30	13	28.4	69.8	2.5	283.1	331.4	12.6
31	11	27.6	66.4	3.8	146.0	333.0	16.6
32	12	13.6	65.2	4.9	80.4	324.3	21.4
33	14	24.9	66.5	2.7	221.0	331.3	17.3
34	12	16.5	68.7	3.2	186.0	324.9	16.2
35	11	24	69.3	2.3	380.7	329.1	13.9
36	11	46.5	71.6	3.3	196.3	338.3	4.5
37	14	37.9	68.3	3.0	173.3	337.4	10.9
38	12	26.0	65.1	3.6	146.3	332.8	18.6
39	11	33.3	77.6	4.8	89.7	327.2	-0.1
40	12	29.1	64.6	4.0	116.1	335.1	18.3
48	12	8.7	66.3	1.9	510.3	320.6	20.6
49	12	25.1	65.4	3.2	181.5	332.0	18.5
50	8	31.0	67.6	2.9	365.5	334.2	14.1
52	13	19.5	65.0	3.1	183.9	328.5	20.5
Component C							
13	5	24.9	59.0	8.8	76.9	335.5	25.6
21	9	4.8	48.5	4.5	134.8	320.1	40.6
22	12	8.3	54.0	3.6	146.8	322.9	35.2
24	10	1.2	62.0	3.7	173.4	315.8	27.3
25	4	8.4	62.0	4.0	107.5	321.6	26.8
26	3	7.8	58.3	8.9	191.4	321.8	31.1
27	3	358.1	41.3	20.5	37.1	312.2	46.8
29	12	4.1	63.4	2.7	267.2	317.8	25.2
30	12	11.7	60.6	2.9	218.2	324.4	27.7
31	3	1.0	61.0	7.8	252.4	315.3	27.7
33	4	2.7	62.4	5.9	241.5	316.6	26.0
34	4	3.7	58.7	3.1	888.9	317.8	30.2
36	12	14.3	60.6	4.5	96.0	326.6	27.3
37	6	18.5	61.0	7.0	93.7	329.6	25.4
38	6	7.1	59.3	6.4	110.5	320.7	29.3
39	3	13.9	65.6	6.6	345.1	324.4	20.8
48	8	355.8	51.2	4.9	129.5	310.0	37.7

The mean direction is given by its declination (*D*), inclination (*I*), radius of cone of 95% confidence (α_{95}), and precision parameter (κ) (Fisher, 1953). VGP is the virtual geomagnetic pole given by paleolongitude (Long.) and paleolatitude (Lat.). *N* is the number of specimens used. The corresponding sites from the sampling localities shown in Fig. 1 are: 7–14 (PM1), 31–35 and 37–52 (PM2), 15–20 (PM3), 24–27 (PM4), 28 (PM5), 21–23, 29–30, and 36 (PM6). This table shows only the sites which gave coherent mean directions.

peratures up to 340°C while the soft fraction demagnetizes up to 500°C heating. The thermal demagnetization applied to other samples from the same site disclosed a B component stable to temperatures up to 340°C [BI52-C2 in Fig. 5(a)]. Above this temperature the magnetization (related to the soft coercivity fraction only) became unstable. These characteristics indicate that the B and C components reliably record an ancient geomagnetic field. However, secular variation of the geomagnetic field seems to have not been fully recorded throughout the sampled profiles, although they may be as thick as 100 m. The calculated angular dispersions (Table 2) are much lower than the predicted values for the present-day field suggesting that secular variation was not

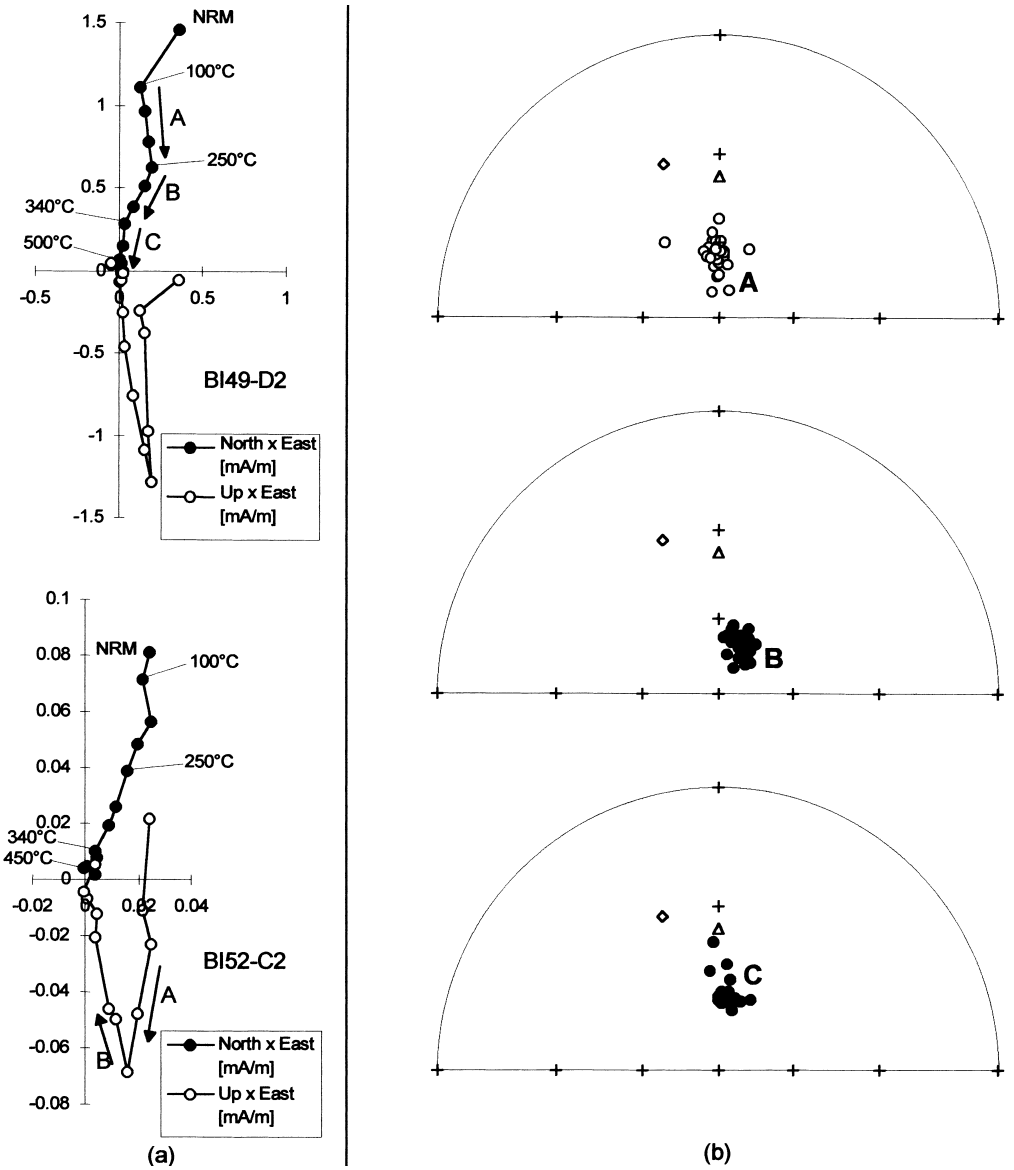


Fig. 5. (a) Orthogonal diagrams (Zijderveld projections) showing the magnetic behavior during thermal demagnetization [open (full) circles are vertical (horizontal) projections]. A, B and C components are identified by the arrows; (b) site mean A, B and C characteristic directions (Wulf projection). ●, positive; ○, negative inclinations. Diamonds and triangles represent the actual geomagnetic field and the dipolar field, respectively.

averaged out. Moreover, samples displayed a single magnetic polarity, which could imply that remanence was acquired in a short time interval.

The inclinations calculated for these components indicate paleolatitudes between 39 and 51° (Table 2). These paleolatitudes would be, at first, opposite to climatic conditions that favor the development of carbonate platforms. According to Ziegler et al. (1984) the development of carbonates is most effective in the zone between 35°N and 35°S, which coincides with the known distribution of Phanerozoic carbonates. Although the par-

Table 2
Neoproterozoic/Lower Paleozoic paleomagnetic poles from the São Francisco Craton[a]

Unit/component	Mean direction							Paleomagnetic pole				
	Dm (°)	Im (°)	α_{95} (°)	κ	N	P (°)	s	Lat. (°N)	Long. (°E)	α_{95} (°)	K	Reference
Bambuí/A	358.0	−64.1	2.3	130.9	31	46.0	6.2	63.9	137.4	3.2	64.6	1
Bambuí/B (BGB)	25.6	68.3	1.6	247.5	33	51.5	5.1	14.7	330.7	2.5	104.3	1
Bambuí/C (BGC)	6.9	58.4	3.2	123.2	17	39.1	7.3	30.2	321.0	3.8	91.2	1
Salitre/SF	3.7	67.1	3.2	126	17	49.8	7.2	27.5	321.4	4.9	55	2
Piquete/PQ	60.4	68.0	6.8	57	9	51.1	10.7	−0.8	346.5	10.2	26	3

[a] N, number of sites; α_{95} (°), K, Fisher's statistical parameters; P, paleolatitude; $s=81/K^{1/2}$, angular dispersion; References: 1, this work [the Bambuí results supersede those from D'Agrella-Filho et al. (1997)]; 2, D'Agrella-Filho (1995); 3, D'Agrella-Filho et al. (1986).

adox of carbonates associated with glacial deposits in the Neoproterozoic could suggest abrupt and severe changes in climatic conditions instead of plate displacement (Fairchild, 1993), the relatively high paleolatitudes inferred for the study area may be associated to quite a different time from that of carbonate sedimentation, and also agree with the post-depositional character of the magnetic components.

The components B and C are similar to that obtained from the Salitre Formation (Una Group) carbonates (Table 2; D'Agrella-Filho, 1995), northern São Francisco basin (Fig. 1), suggesting that the acquisition of these characteristic magnetizations was contemporaneous. The corresponding paleomagnetic poles [after rotation to Africa according to Rowley and Pindell (1989)] are shown in Fig. 6, along with high quality Gondwanan poles for the time interval 550–475 Ma (Meert et al., 1995; Grunow, 1995), considering that by this time the supercontinent was completely assembled. The paleomagnetic pole for the Piquete metamorphic rocks (D'Agrella-Filho et al., 1986), southern São Francisco craton, is also shown. The Piquete rocks yield typical Brasiliano K–Ar ages in the range 531.7 ± 13.5 to 467 ± 15.3 Ma, with a peak in the 500–490 Ma interval. All the São Francisco craton poles plot very close to the 522 ± 13 Ma Ntonya Ring (NR) pole, the ca. 515 Ma Sør Rondane Intrusions pole and the 510 ± 15 Ma Central Australia mean pole, suggesting that the characteristic magnetization obtained in the southern (Bambuí Group) and northern (Salitre Formation) parts of the basin, as well as in the surrounding metamorphic rocks (Piquete Complex), were acquired at 530–500 Ma, during the last stage of the Brasiliano orogeny.

5. Pb–Pb and U–Pb geochronology and Pb isotope geochemistry

The isotopic analyses were carried out at the Isotope Geochemistry Laboratory of the University of Kansas, and all the analytical procedures are described in detail by Babinski et al. (1999). The isochrons were regressed using the ISOPLOT program of Ludwig (1990).

Based on the Pb isotopic compositions and U/Pb ratios of the analyzed rocks, four types of Pb were determined on the carbonates and classified as Types I, II, III and IV (Table 3.; Babinski et al., 1999). Type I Pb was found in samples with low Pb concentrations (0.10–0.88 ppm) and relatively high U concentrations (0.17–0.95 ppm) showing U/Pb ratios generally >1; it represents in situ growth of radiogenic Pb and is able to yield reliable Pb–Pb isochron ages. Type II Pb is present in samples with relatively high Pb concentrations (0.76 to 35.0 ppm) and low U concentration (0.01–3.9 ppm); it is non-radiogenic crustal Pb. Type III Pb is also found in samples with high Pb concentrations (1.8–50.0 ppm) and low U concentrations (0.19–1.22 ppm) but it is radiogenic crustal Pb derived from the Archean/Paleoproterozoic basement. Type IV Pb occurs in samples with U/Pb ratios <1 and is intermediate in composition between Types III and I Pb; it represents a mixture

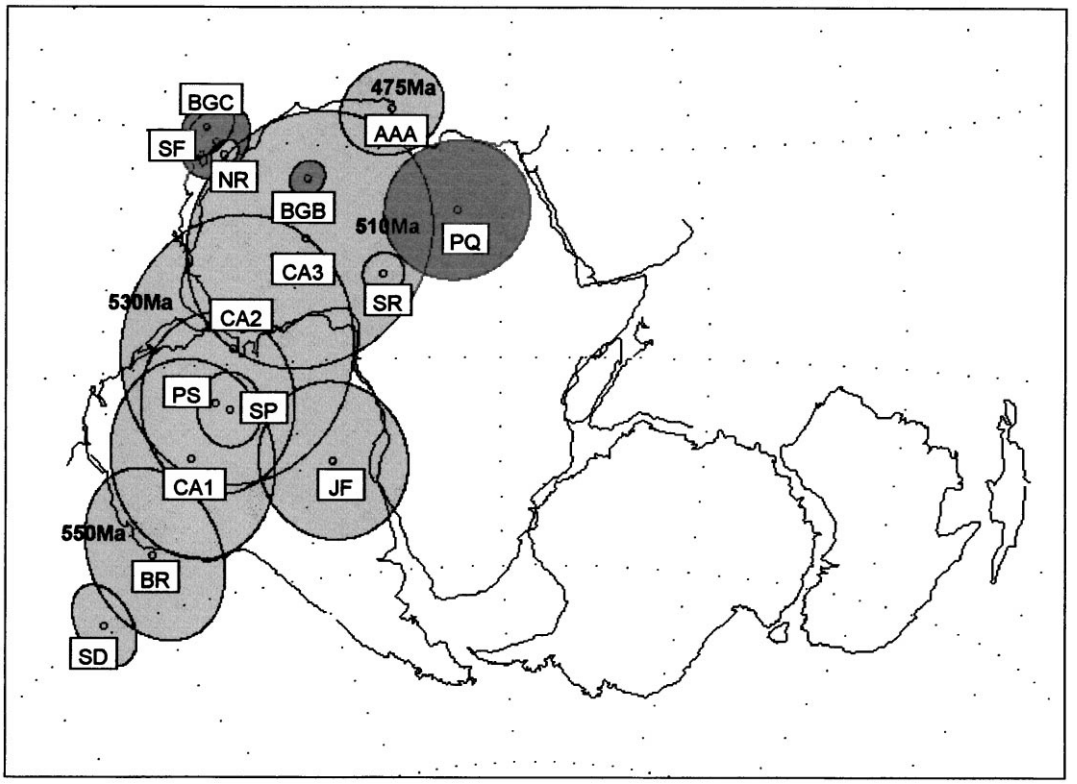


Fig. 6. 550–475 Ma Gondwana selected poles. Bambuí Group (BGB, BGC): this work; Salitre Formation (SF): D'Agrella-Filho (1995); Piquete Complex (PQ): D'Agrella-Filho et al. (1986); Sinyai Dolerite (SD), Ntonya Ring Structures (NR), Bhander and Rewa (BR), Purple Sandstone (PS), Jutana Formation (JF); Salt Pseudomorph's (SP), Sør Rondane Intrusions (SR), Central Australia mean poles (CA1, CA2, CA3): Meert et al. (1995); Australia, Africa and Antarctica 475 Ma mean pole (AAA): Grunow (1995). São Francisco Craton poles (dark gray) and South America restored to Africa (in its present position) according to reconstruction of Rowley and Pindell (1989) (rotation pole 47.23°N, 30.84°E, 55.69° for South America to Africa). Equal-area projection.

of those two types. Although most outcrops contain only one type of Pb, three out of four types of Pb (I, III and IV types) were found in rocks of outcrop MF-7. The Pb isotopic compositions, the Pb and U concentrations data as well as the detailed sample collection and description of the rocks used in this work is given in Babinski et al. (1999).

Table 3
General characteristics of the different types of Pb determined on the Bambuí Group carbonate rocks[a]

Type of Pb	$^{206}Pb/^{204}Pb$	$^{207}Pb/^{204}Pb$	Pb (ppm)	U (ppm)	U/Pb
Type I Pb	18.91–80.90	15.68–19.42	0.10–0.88 (3.9)	0.17–0.95	>1.0
Type II Pb	18.23–19.71	15.65–15.76	0.76–35.0 (164.6)	0.01–3.09	0.02–0.1
Type IIIa Pb	35.62–36.63	18.26–18.48	4.22–50.0 (455.5)	0.19–1.22	0.07–0.1
Type IIIb Pb	32.42–34.28	17.65–17.99	2.11–9.98	0.28–0.57	~0.1
Type IIIc Pb	30.33–31.15	17.32–17.36	1.81–3.26	0.34–0.38	0.1–0.2
Type IV Pb	24.12–45.35	16.35–18.59	0.12–0.74	0.14–0.67	<1.0

[a] Numbers in parentheses represent extreme values of the concentrations obtained for that type of Pb.

Only four out of eight outcrops showed Type I Pb, the only type which is able to yield isochron ages. Samples from the other outcrops contain other types of Pb which can yield information about the behavior of the Pb isotopes during post depositional events that took place in the basin.

Two outcrops (MF-7 and MF-10) located within the undeformed area contain Type I Pb. Data from the MF-7 samples showed a ^{207}Pb–^{206}Pb isochron age of 686 ± 69 Ma [Fig. 7(a)]; data from the same rocks plotted on a $^{206}Pb/^{204}Pb$ versus $^{238}U/^{204}Pb$ diagram show large scatter with no alignment [Fig. 7(b)] indicating that the rocks did not behave as closed U–Pb system. Since good alignment of Pb isotope ratios was obtained, we believe that the scatter observed on the U–Pb diagram was caused by recent U lost. Samples from the MF-10 outcrop yielded a ^{207}Pb–^{206}Pb age of 520 ± 53 Ma [Fig. 8(a)]; on a $^{206}Pb/^{204}Pb$ versus $^{238}U/^{204}Pb$ diagram, although some scatter is observed, a slope that indicates an ^{238}U–^{206}Pb age of 603 ± 80 Ma [Fig. 8(b)] can be determined for these rocks.

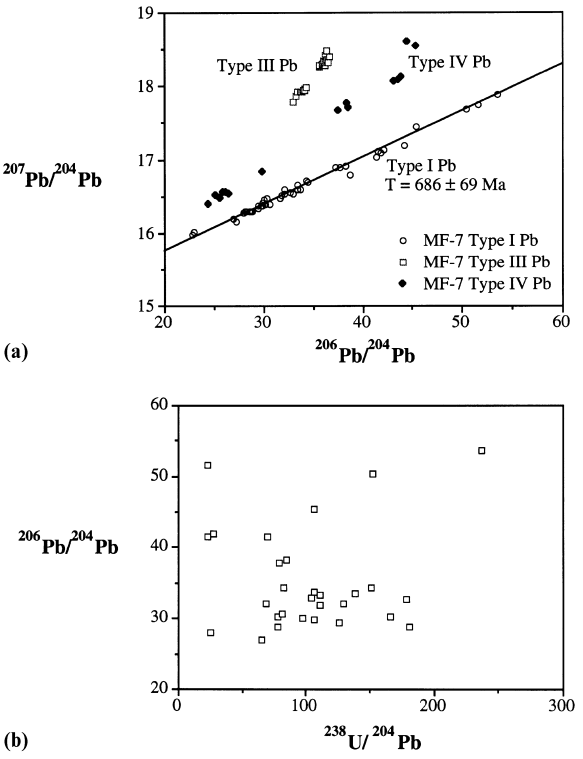


Fig. 7. (a) Pb diagram showing the three different types of Pb determined on the MF-7 samples. The age of 686 ± 69 Ma was determined from samples containing Type I Pb. (b) U–Pb diagram from samples containing Type I Pb.

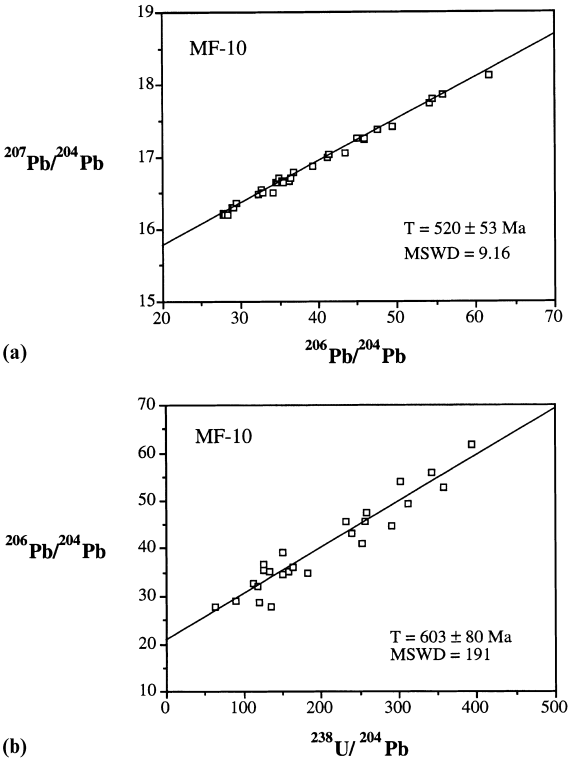
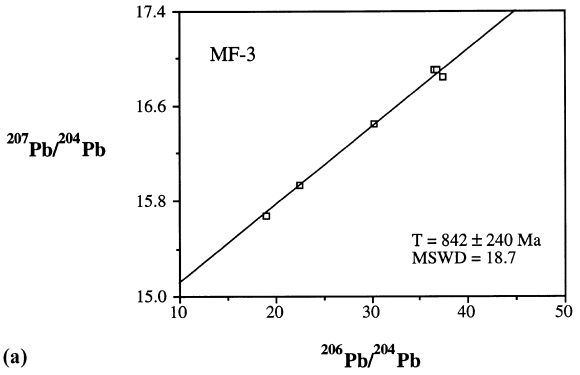


Fig. 8. (a) Pb diagram from samples MF-10. (b) U–Pb diagram from samples MF-10.

Type I Pb was also found in carbonates collected from areas affected by the Brasiliano deformation. MF-3 samples define a ^{207}Pb–^{206}Pb age of 842 ± 240 Ma [Fig. 9(a)]; $^{238}U/^{204}Pb$ ratios plotted against $^{206}Pb/^{204}Pb$ isotopic compositions give an ^{238}U–^{206}Pb errochron age of 545 ± 210 Ma [Fig. 9(b)]. In this case some scatter is also observed, but a preferential trend showing a slope that yields an age of ca. 550 Ma is obtained. Carbonates from MF-9 outcrop yielded a ^{207}Pb–^{206}Pb age of 872 ± 290 Ma [Fig. 10(a)] and an ^{238}U–^{206}Pb age of 621 ± 160 Ma [Fig. 10(b)].

Type II Pb was determined on samples from three outcrops: one is located on the stable area of the basin (MF-6) and the others (MF-5 and MF-11) are from the area affected by the Araçuaí



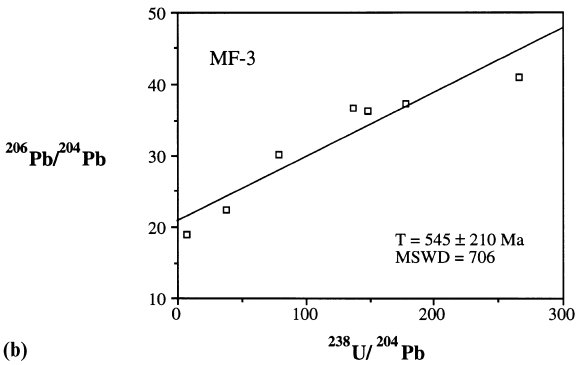
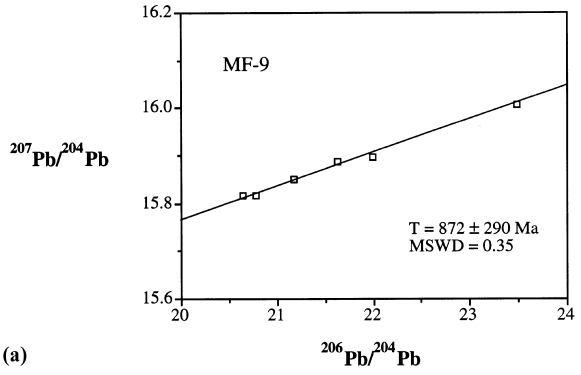


Fig. 9. (a) Pb diagram from samples MF-3. (b) U–Pb diagram from samples MF-3.



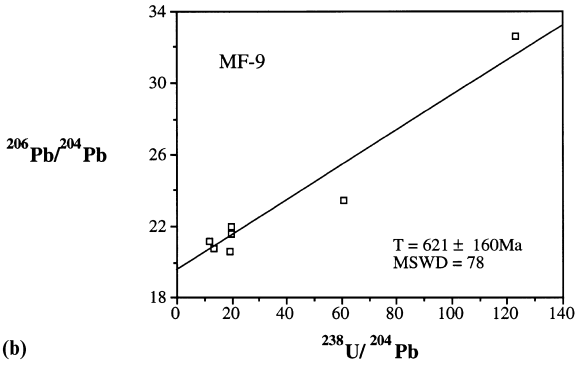


Fig. 10. (a) Pb diagram from samples MF-9. (b) U–Pb diagram from samples MF-9.

fold belt. Their Pb isotopic compositions are non-radiogenic ($^{206}Pb/^{204}Pb = 18.8$ and $^{207}Pb/^{204}Pb = 15.75$), representing average crustal Pb, and they fall slightly above the Stacey and Kramer's (1975) Pb evolution curve. This type of Pb does not provide isochron ages.

Type III Pb is present in carbonates collected in two outcrops (MF-7 and MF-17) located in the undeformed area (Fig. 1). Its Pb isotopic compositions represent radiogenic crustal Pb (Table 3); they define a straight line and its slope indicates an *apparent* ^{207}Pb–^{206}Pb age of ca. 2.5 Ga, which could represent the age of the source of this 'detrital Pb' that was later incorporated into the Neoproterozoic carbonates.

Type IV Pb was found in carbonates from the MF-7 outcrop where types I and III Pb were also determined. The samples that contain Type IV Pb were collected from intermediate positions between samples with types I and III Pb, and their Pb isotopic compositions represent the mixture of those two types (Table 3).

The Pb–Pb isochron ages obtained from mesoscopically undeformed carbonates containing Type I Pb are 686 ± 69 Ma (MF-7) and 520 ± 53 Ma (MF-10). The ^{238}U–^{206}Pb age determined for the MF-10 outcrop is 603 ± 80 Ma; MF-7 samples do not show any alignment on the U–Pb diagram. Because the older age (686 ± 69 Ma) was determined on the same outcrop (MF-7) where 'detrital' Pb was also detected (Types III and IV, which were presumably incorporated into the carbonates during the Brasiliano orogeny), this age is considered as the minimum depositional age for the carbonates from the Sete Lagoas Formation (Bambuí Group basal unit). The younger age (520 ± 53 Ma) was determined on carbonates which are stratigraphically below outcrop MF-7, so that this age should represent the time of the re-homogenization of the isotopic system during a

post-depositional event. Carbonates showing deformation (MF-3 and MF-9) yielded Pb–Pb ages of 842 ± 240 Ma and 872 ± 290 Ma, and ^{238}U–^{206}Pb ages of 545 ± 210 and 621 ± 160 Ma, respectively. Although the ages present large errors, the U–Pb ages also point to an isotopic resetting at ca. 550–500 Ma. This resetting event is shown by both deformed and underformed rocks and could have been promoted by percolation of fluids generated during the Brasiliano orogeny.

Type II Pb has isotopic compositions that could represent average seawater composition at the time of deposition of the carbonates. However, since the same isotopic pattern was determined on undeformed as well as in samples clearly affected by the tectonism (marbles), we prefer to interpret them as the Pb isotopic composition acquired by the rocks during the Brasiliano orogeny.

Isotopic compositions from Type III Pb (old Pb) can be divided in three groups (Types IIIa, IIIb and IIIc). These three groups were evolved from different high-μ domains. Because these samples do not show mesoscopic deformation, although the rocks are recrystallized, we suggest that the old Pb was incorporated in the Neoproterozoic carbonates through a large scale fluid percolation process.

The same Pb isotopic signature of Type III Pb was determined on galenas from the western part of the basin. In Fig. 11, we present the Pb isotopic compositions determined on the galenas [data summarized by Iyer et al. (1992)] and those obtained on the carbonates hosting Type III Pb, showing that only one trend is defined by them. The galenas were formed during one or more pulses of the Brasiliano orogeny, and the source of Pb is defined as the basement rocks (Parenti-Couto et al., 1981). In the same figure we also plotted the Pb isotopic compositions determined on samples hosting Type II Pb, that is, non-radiogenic crustal Pb. As can be observed, the average Pb isotopic compositions of Type II Pb fall at the lower end of the trend defined by samples hosting Type III Pb and by the galenas. This suggests that the isotopic compositions of Type II Pb were generated during the same event that incorporated Type III Pb in the carbonates and formed the galenas, where favorable structural and chemical traps existed.

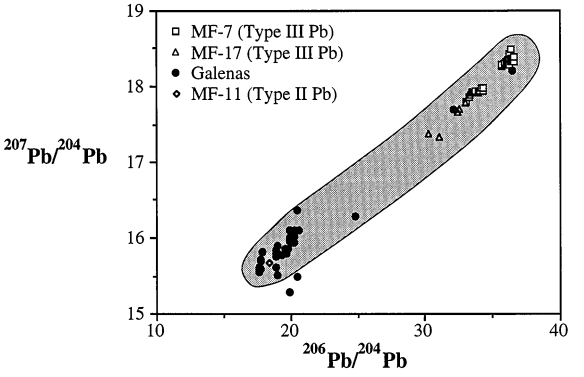


Fig. 11. Pb diagram showing the isotopic compositions determined for carbonate samples containing Type III Pb (Pb from the basement) and Pb data obtained for galenas within Bambuí carbonates [summarized by Iyer et al. (1992)], indicating that the Type III and galena Pb came from the same or similar sources. Pb isotopic compositions of Type II Pb are plotted for reference.

Assuming that Types III and II Pb were introduced into the carbonates at the same deformational event, a regression of their Pb isotopic ratios was done (Fig. 12) and it intercepts the Stacey and Kramer's (1975) Pb evolution growth curve at ca. 520 and 2100 Ma. This line is interpreted as an array of Pb isotopic compositions generated

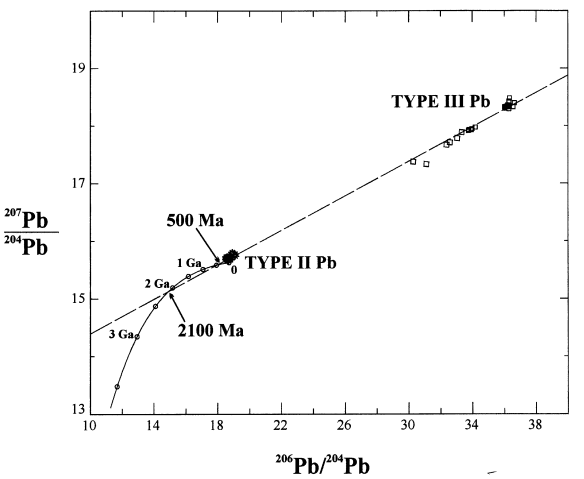


Fig. 12. Pb diagram showing the isotopic compositions determined on samples presenting Types II and III Pb. These data define a straight line that intercepts the Stacey and Kramers (1975) Pb evolution curve at 520 and 2100 Ma. This Pb probably represents crustal Pb from variable U/Pb domains in basement rocks of the São Francisco basin at 500–550 Ma.

during a third stage of Pb evolution with variable μ, indicating that old Pb from the basement was incorporated into the carbonate rocks at about 500–550 Ma through a large scale fluid percolation event that could have begun before 500–550 Ma. However, it is not possible to define, with the available isotopic data, the precise time that the process started. In any case, the 500–550 Ma could be either the time of the stronger pulse of fluid percolation or the time that the process stopped, and since then the U–Pb isotopic system was not longer disturbed yielding Pb–Pb and U–Pb isochrons ages of ca. 500–600 Ma on the carbonates.

6. Conclusions

Rock magnetism, paleomagnetism and U–Pb and Pb isotope data give independent evidence for a strong event of remagnetization and resetting of the isotopic system in the last stage of the Brasiliano/Pan-African orogeny.

The post-depositional character of the characteristic magnetizations is strengthened by the following:

1. the rock magnetic properties are typical of remagnetized carbonates found in other parts of the world;
2. moderate to high paleolatitudes inferred from paleomagnetic data for the study area would require a different climate pattern during the sedimentation of the wide carbonate platforms;
3. magnetization directions with a single polarity were found for >100 m in the sedimentary sequence;
4. mean magnetization components identified in the Bambuí and Una (Salitre Formation) carbonates show lower dispersion than would be expected if the secular variation of the geomagnetic field was fully averaged out; and
5. paleomagnetic poles and radiometric ages from carbonate sequences and adjacent Brasiliano metamorphic rocks are coincident.

Fluid inclusion studies done in the fluorites and willemites (Dardenne and Freitas-Silva, 1998) from the carbonates indicate temperatures between 100 and 200°C for the fluids that percolated in the basin. These studies also showed hydrocarbon fluids and solid bitumen in some of the fluid inclusions. Previous analyses of thermal alteration of palinomorphs incorporated in the sedimentary rocks (equivalent to the Vitrinite Reflection Index), and carbon isotopic compositions of methane recovered from the central area of the basin suggested that the rocks were not submitted to temperatures >200°C (Babinski et al., 1989). These results together with the authigenic magnetites observed in the SEM images and the rock magnetic properties support a chemical remagnetization due to percolation of hydrocarbon-bearing fluids (e.g. McCabe and Elmore, 1989).

Pb–Pb and the U–Pb ages obtained in the deformed as well as in undeformed carbonates fall in the same range (considering the analytical errors). Because of that we interpret these ages as post-depositional ages. Furthermore, carbonates containing Pb Type III ('detrital' Pb) are found in the central portion of the basin which was not submitted to deformation and high temperatures. This indicates that Pb Type III, derived from the Archean/Paleoproterozoic basement, was incorporated through fluids which also formed the galenas and promoted a severe remagnetization in the carbonates. The trend defined by Pb isotopic compositions of Pb Types III and II intercepts the Stacey and Kramer's Pb evolution curve at ca. 2100 and 520 Ma, suggesting that the fluid percolation event took place at ca. 500–550 Ma. The paleomagnetic results permit further constraint on these ages to the 530–500 Ma interval by correlations with high quality paleomagnetic poles in the 550–475 Ma Gondwana APWP (Fig. 6).

The similarity between paleomagnetic results from the Bambuí and the ca. 1000 km far Salitre Formation (Una Group) carbonates implies in a large scale fluid percolation event that simultaneously affected the whole basin. Preliminary Pb isotope results from carbonates of the northern part of the basin corroborate this hypothesis. The paths of these fluids may have been along old basement faults reactivated during the last stage of the Brasiliano orogeny.

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